

Saumil Singhal

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EDUCATION

Vellore Institute of Technology, Bhopal

Bhopal, India

B.Tech in Computer Science Engineering (Artificial Intelligence & Machine Learning)

2023 – 2027

Relevant Coursework: Data Structures & Algorithms, Operating Systems, Computer Networks, Database Management Systems, Cloud Computing

EXPERIENCE

Bharat Heavy Electricals Limited (BHEL)

Haridwar, India

Full-Stack Developer Intern

May 2026 – June 2026

- Designed a **relational database schema (SQLite/PostgreSQL)** and developed a scalable **Node.js/Express** backend with **React** frontend for the Smart Material & Inventory Management Platform (SMIMP), building production-ready **REST APIs** and backend workflows.
- Implemented a **predictive forecasting engine** to automate material demand analysis, exposing secure REST API endpoints and optimizing backend data processing workflows.
- Owned end-to-end development including database design, API development, debugging, deployment, testing, Git-based version control, and collaboration with stakeholders to deliver reliable production-ready features.

PROJECTS

AI-Powered Finance Management Platform | *Next.js, React, Prisma, Supabase, Tailwind CSS* Dec 2025 – Jan 2026

- Built and deployed a scalable **AI-powered** personal finance platform with authentication, PostgreSQL, REST APIs, and modular backend workflows to help users manage spending and budgets.
- Designed relational database schemas and implemented secure authentication, backend services, and API-driven workflows using **Prisma ORM** and **Supabase PostgreSQL**.
- Engineered AI-powered financial insights by processing structured financial datasets through scalable backend services and intelligent recommendation workflows.

Credit Card Approval Model | *Python, TensorFlow, React, Flask, Scikit-learn*

May 2025 – Jun 2025

[\[GitHub\]](#) [\[Live Demo\]](#)

- Trained and evaluated multiple machine learning models on financial applicant data to automate credit risk prediction and identify key approval factors.
- Developed and deployed **RESTful backend services** using Flask, enabling scalable API-driven communication between the prediction engine and React frontend.
- Delivered a live production-ready application with a publicly accessible GitHub repository, demonstrating end-to-end software engineering and deployment practices.

Stock Market Trend Predictor | *Python, TensorFlow, Pandas, NumPy, Keras*

Apr 2025 – Jun 2025

[\[GitHub\]](#) [\[Live Demo\]](#)

- Built an end-to-end forecasting system using **LSTM** models on live **yFinance API** data to predict market trends.
- Experimented with multiple forecasting architectures, evaluating performance against historical market data.
- Deployed an interactive **Streamlit** dashboard, shipping the model as a usable, live analytics tool.

SKILLS

Languages: Java, Python, JavaScript, SQL

Computer Science Fundamentals: Data Structures & Algorithms, Object-Oriented Programming (OOP), Computer Networks, Operating Systems, Database Management Systems (DBMS), Complexity Analysis

Cloud & DevOps: Cloud Computing, Cloud Operations, Infrastructure Automation, Infrastructure as Code (IaC), CI/CD, Linux, Docker (Basics), Kubernetes (Basics), AWS (Basics), Google Cloud Platform (Basics), Site Reliability Engineering (SRE), Monitoring, Logging, Observability, Metrics, Configuration Management

Backend Development: Node.js, Express.js, Flask, FastAPI, REST APIs, Authentication, API Design, Backend Development, Microservices

Databases: PostgreSQL, SQLite, MongoDB, Prisma ORM, SQL, NoSQL

AI & Data Science: AI-assisted Development (Claude, ChatGPT), Prompt Engineering, LLM Evaluation, Error Analysis, Predictive Modeling, TensorFlow, Keras, Scikit-learn, Pandas, NumPy

Developer Tools: Git, GitHub, GitHub Actions, Postman, VS Code, Streamlit

CERTIFICATIONS

Java Foundation Associate – Oracle 

Applied Machine Learning in Python – Coursera 